Lab 6: Linear Models

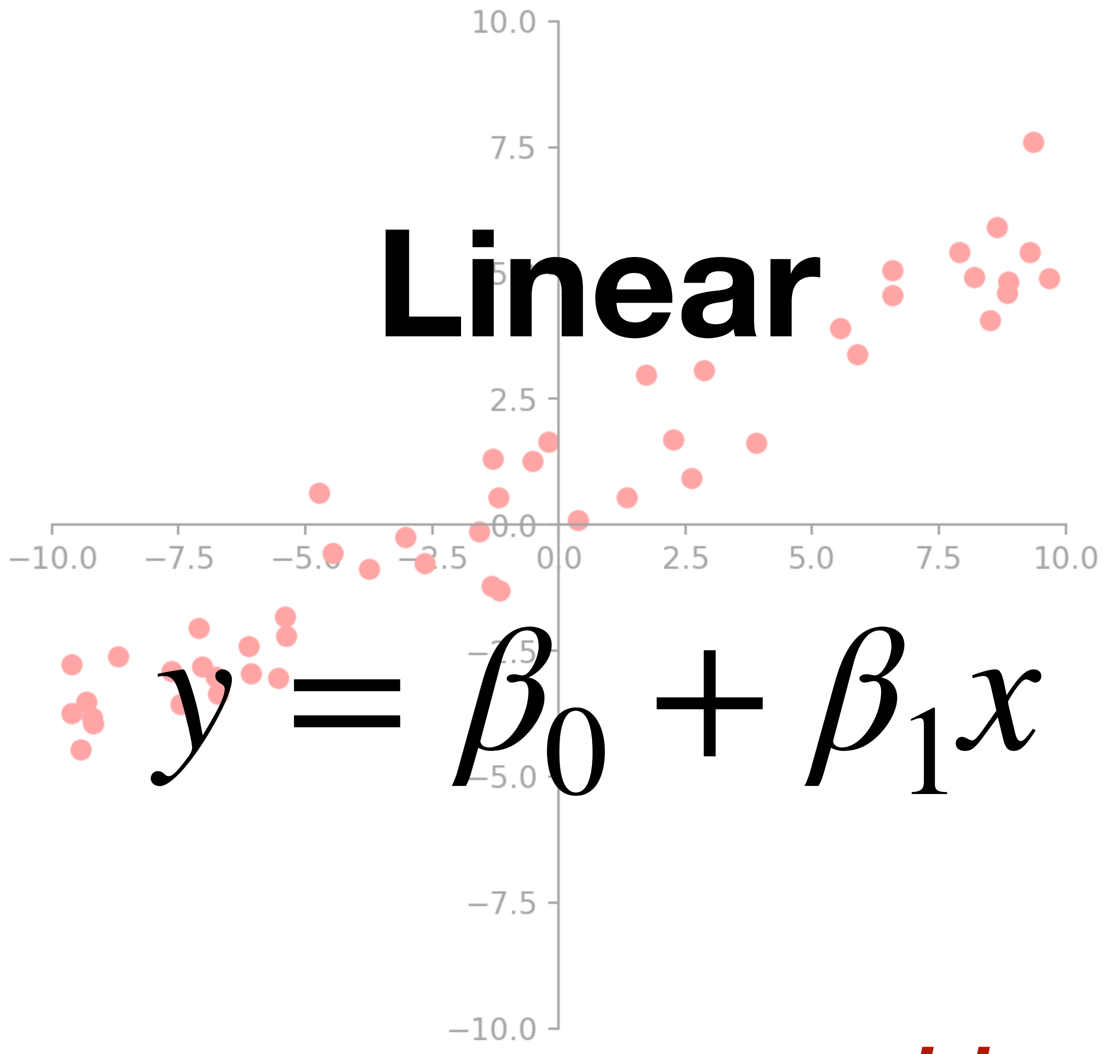
CAS CS 132: Geometric Algorithms

Teaching Fellow: Angelos Poulis (apoulis@bu.edu)
April 22nd, 2026

What is a Model?

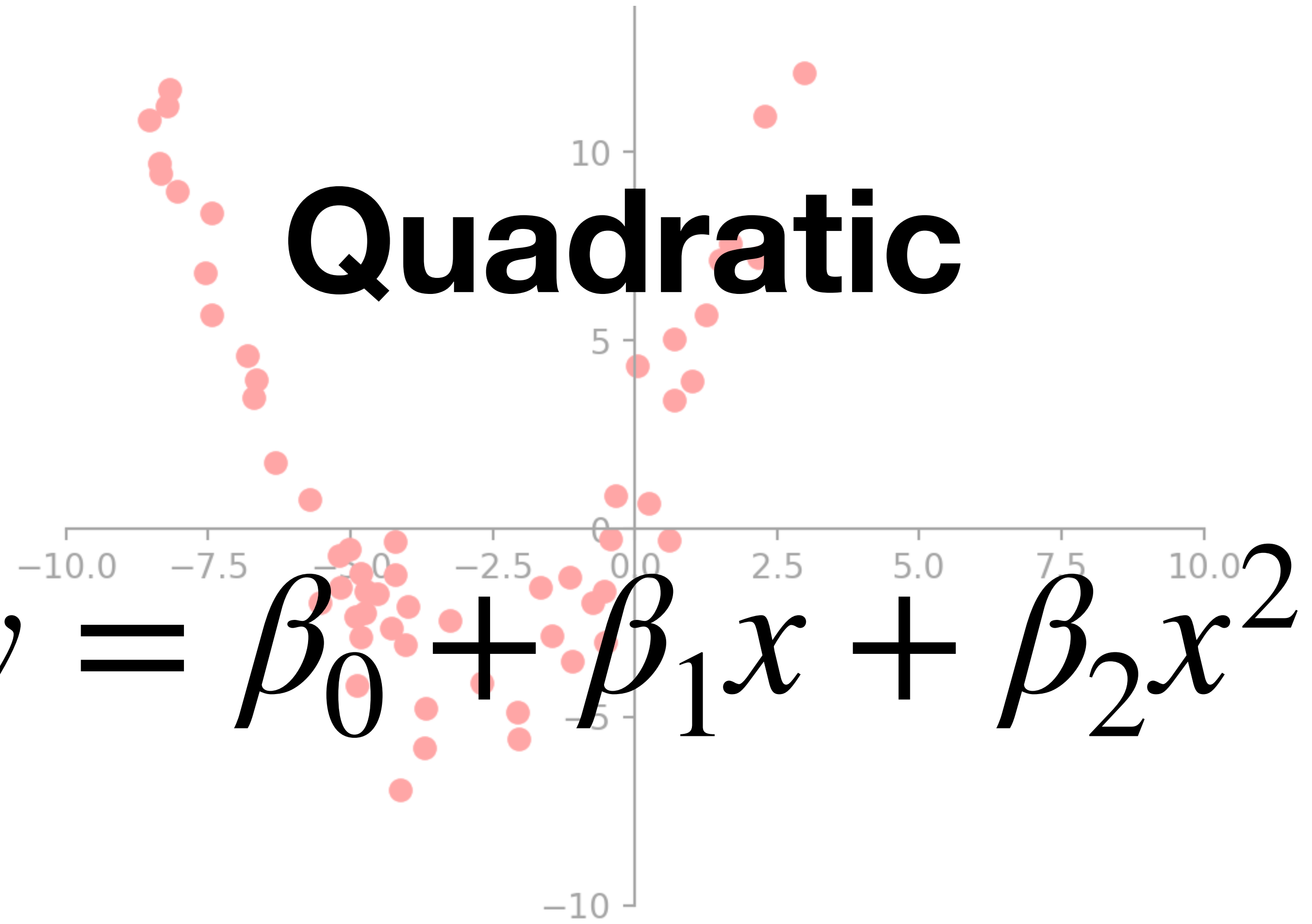
A concise description of a dataset or a real-world phenomenon.
An equation can be a model if we use it to describe something in the real-world.

The most common form of modeling is **regression**: constructing an equation that describes the relationships among variables.



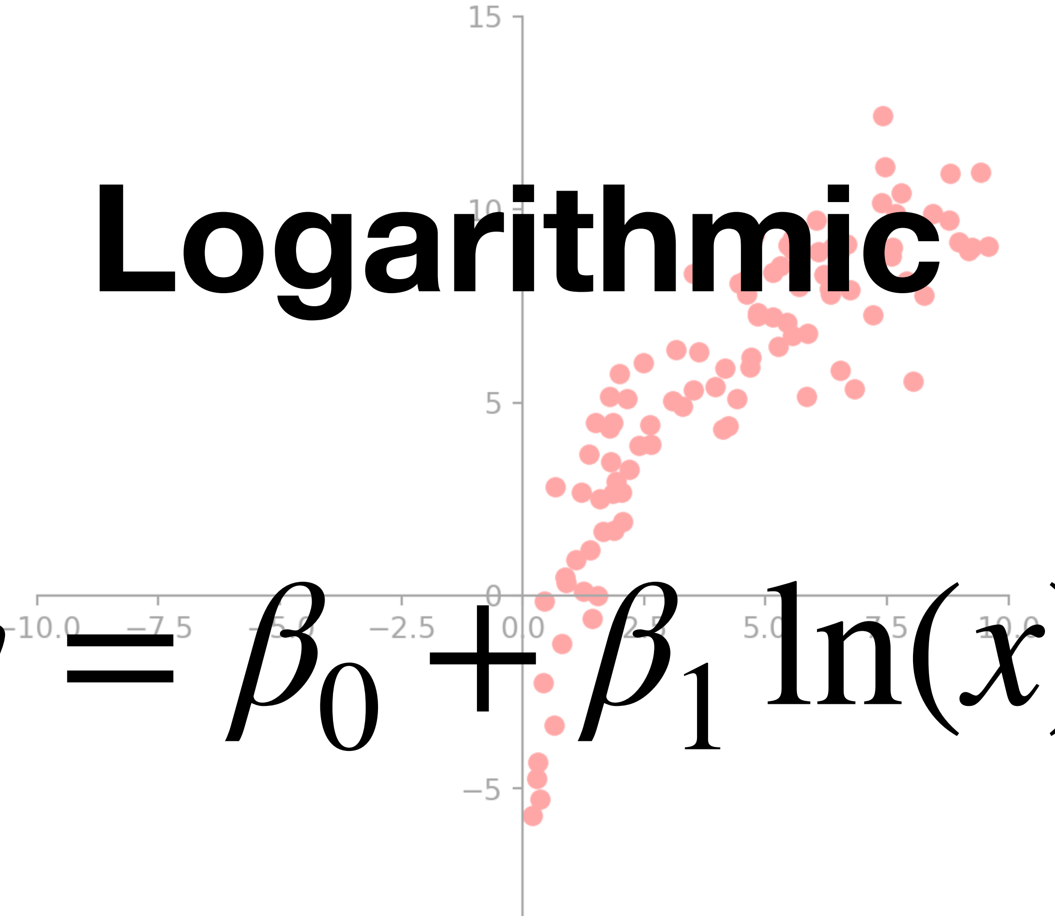
Linear

$$y = \beta_0 + \beta_1 x$$



Quadratic

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$



Logarithmic

$$y = \beta_0 + \beta_1 \ln(x)$$

*How do we find the **best** model given the data?*

Regression Setup

- **Independent variables:** inputs/features
- **Dependent variables:** the output we want to predict
- **Design matrix $\mathbf{X}$:** rows = data points; cols = features
- **Observation vector $\mathbf{y}$:** the dependent variables
- **Parameter vector β:** the unknowns we solve for

Goal 🎯

Find β s.t.

$$\mathbf{X}\beta \approx \mathbf{y}$$

Task: Given a set of independent variables, and the associated dependent variables, estimate the parameters of a model that describes how the dependent variables are related to the independent variables.

Fitting a Line to the Data

The simplest model is a **line**.

If data fell *exactly* on the line, each point would satisfy:

$$
\begin{aligned}
y_1 &= \beta_0 + \beta_1 x_1 \\
y_2 &= \beta_0 + \beta_1 x_2 \\
&\cdots \\
y_n &= \beta_0 + \beta_1 x_n
\end{aligned}
$$

$$\mathbf{y} = \mathbf{X}\beta$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

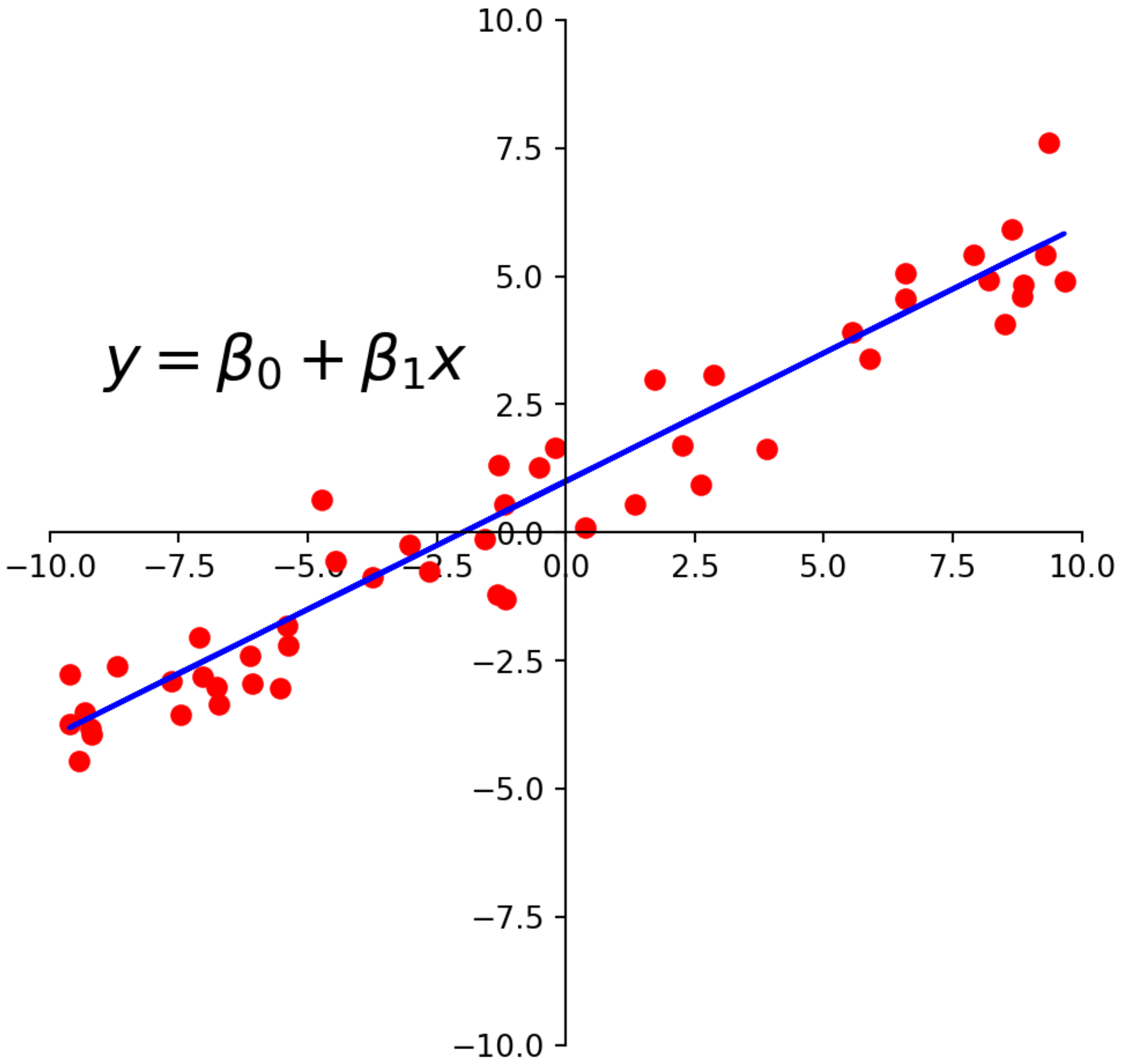
Least Squares

Data rarely falls exactly on a line, so $\mathbf{X}\beta = \mathbf{y}$ has **no solution**.

Instead, we try to best approximate $\mathbf{X}\beta \approx \mathbf{y}$.

That is, minimize the sum of squared errors:

$$\|X\boldsymbol{\beta} - \mathbf{y}\|^2$$



Normal Equations: $X^\top X\boldsymbol{\beta} = X^\top \mathbf{y} \implies \hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$

"Linear in Parameters"

A model is ***linear in its parameters*** (β) when each observation (y) is a linear combination of known functions:

$$y = \beta_0 f_0(x) + \beta_1 f_1(x) + \ldots + \beta_n f_n(x),$$

Where $f_0, \ldots, f_n$ are known functions.

E.g., $y = \beta_0 + \beta_1 x + \beta_2 x^2$ **is linear in β,** quadratic in x

E.g., $y = \beta_0 e^{-\beta_1 x}$ is ***not*** linear in β

Multiple Regression

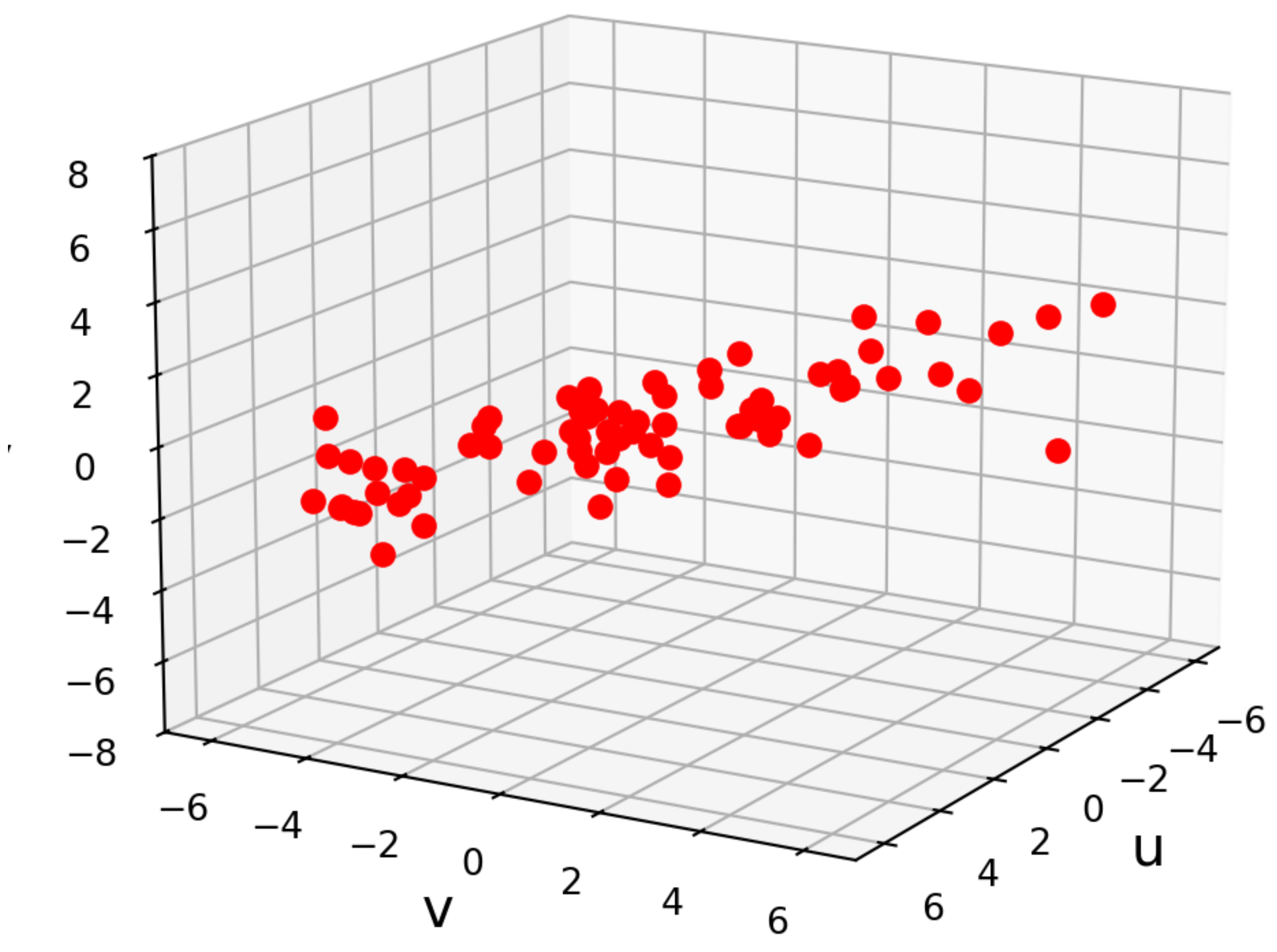
When there is more than one independent variable, we have multiple regression.

Linear: $y = \beta_0 + \beta_1 u + \beta_2 v$

Trend surface:

$$y = \beta_0 + \beta_1 u + \beta_2 v + \beta_3 u^2 + \beta_4 uv + \beta_5 v^2$$

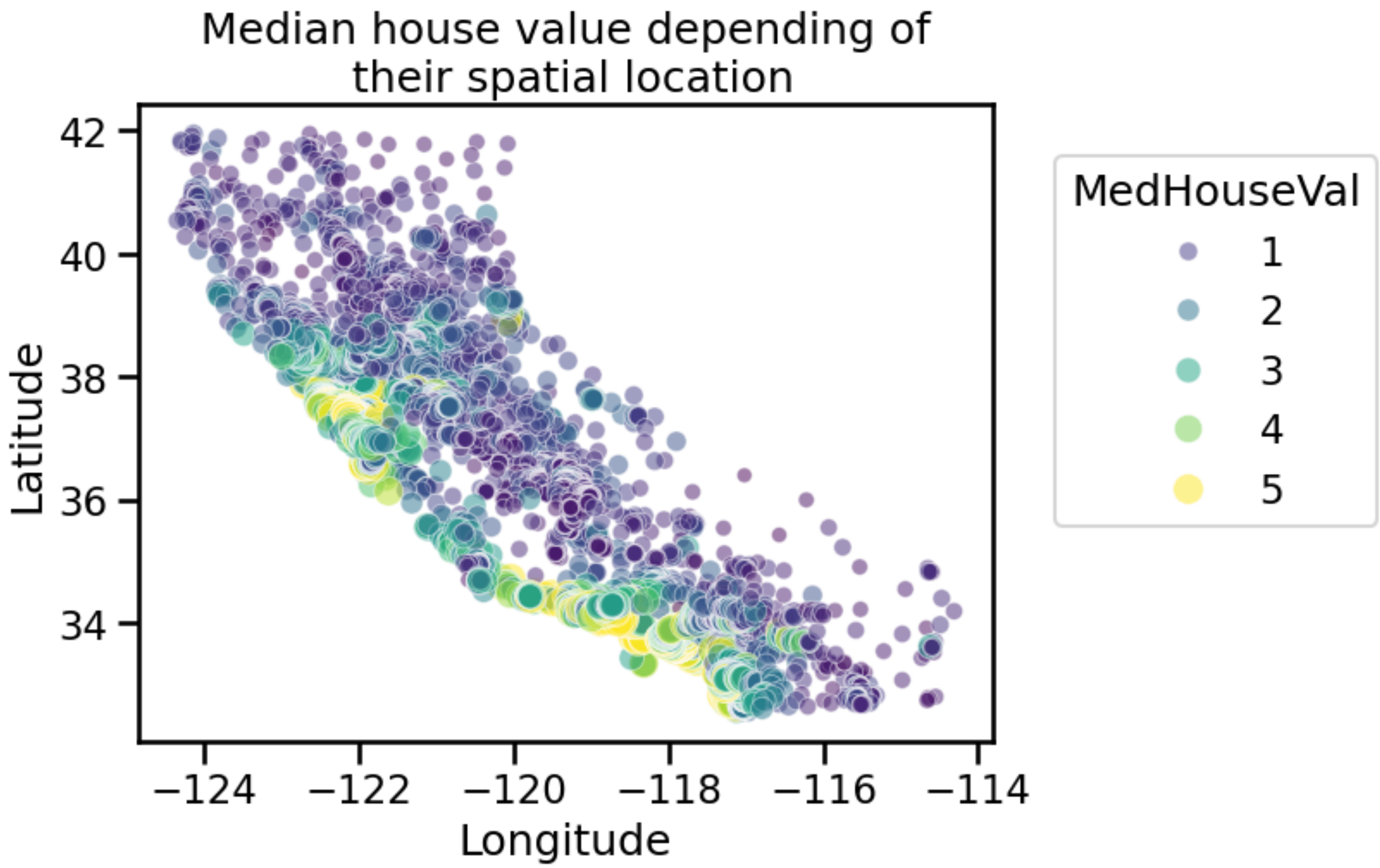
Terrain Data for Multiple Regression

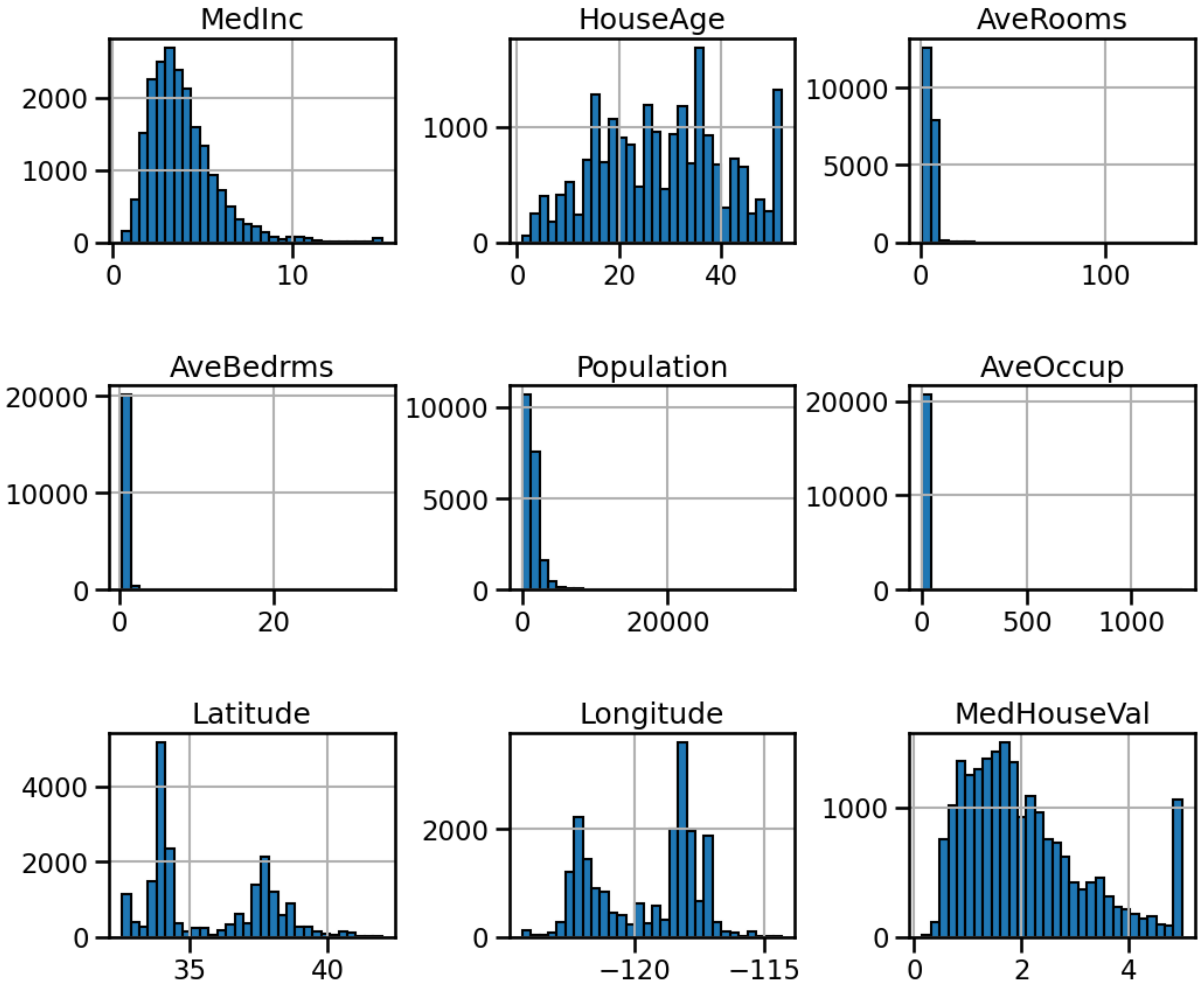


Key principle: Once $\mathbf{X}$ is defined properly, the normal equations have the same matrix form no matter how many variables are involved. For any linear model where $\mathbf{X}^\top\mathbf{X}$ is invertible, $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$

The Lab: California Housing 🏡

Predict median house values in California districts using 1990 census data.





Ref: https://inria.github.io/scikit-learn-mooc/python_scripts/datasets_california_housing.html

- **Link to Lab 6:** https://nmmull.github.io/cs132-s26/labs/6/lab6.pdf
- **Reading:** https://mcrovella.github.io/CS132-Geometric-Algorithms/L23LinearModels.html
- **Write-up due:** April 30, 2026 by 8:00PM